145. A hollow sphere of radius R, a hollow cube of side R and a thin circular plate of radius R, made up of the same material, are all heated to 20° C above room temperature. When left to cool in the room, which of them will reach the room temperature first?

(a) Circular plate
(b) Cube
(c) Sphere
(d) All of them will reach the room temperature at the same time

Ans. (c) : The loss of heat is directly proportional to the surface area, hence loss of heat will be fast in case of sphere because it has the maximum surface area.

146. The age of a man is three times the sum of the ages of his two sons. Five years hence, his age will be double of the sum of the ages of his sons. The father's present age is

(a) 40 years (b) 45 years
(c) 50 years (d) 55 years

Ans. (b) : Let the sum of ages of two sons be x years
$\therefore$ Age of the man = $3x$ years
After 5 years age of man = $(3x + 5)$
After 5 years age of two sons combined = $(x + 10)$ years
Given $(3x + 5) = 2x + 20$
$x = 15$
Hence $3x = 45$
$\therefore$ Current age of father is 45 years.

147. The length of the longest pole than can be placed in a room 12m long and 8 m wide and 9 m high is:

(a) 12 m (b) 14 m
(c) 17 m (d) 21 m

Ans. (c) : Length of the longest pole = length of the diagonal of the room $= \sqrt{(12)^2 + (8)^2 + (9)^2}$
$= \sqrt{289} = 17$m.

148. In a company, 60% of the employees are men. Of these 40% are drawing more than Rs. 50,000 per years. If 36% of the total employees of the company draw more than Rs. 50,000 per year, what is the percentage among women who are drawing less than Rs. 50,000 per year?

(a) 70 (b) 60
(c) 40 (d) 30

Ans. (a) : Let total no. of employees be 100
$\therefore$ No. of men $= \frac{60 \times 100}{100} = 60$
No. of women $= \frac{40 \times 100}{100} = 40$
No. of men drawing more than Rs. 50000
$\frac{40 \times 60}{100} = 24$
Since no. of total employees drawing more than
$5000 = \frac{36 \times 100}{100} = 36$
No. of women who draw more than 50000
$= 36 - 24 = 12$
$\therefore$ No. of women who draw less than 5000
$= 40 - 12 = 28$
$\therefore$ Percentage of such women
$= \frac{28 \times 100}{40} = 70\%$

149. Consider the following statements:

1. **Light of longer wavelength is scattered much more than the light of shorter wavelength.**
2. **The speed of visible light in water is 0.95 times the speed in vacuum.**
3. **Radio waves are produced by rapidly oscillating electrical currents.**
4. **To detect the over speeding vehicles, police use the Doppler effect of reflected short radio waves.**

Which of these statements are correct?

(a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 3 and 4

Ans. (d) : Light of longer wavelength (lower refractive index) is scattered less than light of shorter wavelength. Speed of visible light in water = 2.25 x 10^8 M/Sec. And the speed of light in vacuum is 3.0 x 10^8 M/Sec. It is clear that the speed of visible light in water is 0.75 times the speed of light in vacuum. Thus statement 1 and 2 are false and statement 3 and 4 are correct.

150. A bus is moving with a speed of 30 km/hr ahead of a car with a speed of 50 km/hr. How many kilo meters apart are they if it takes 15 minutes for the car to catch up with the bus?

(a) 5 km (b) 7.5 km
(c) 12.5 km (d) 15 km

Ans. (a) : According to question relative speed of the car with respect to the bus = 50 – 30 = 20 km/hr.
Now distance between the car and the bus, when care catches the bus = distance travelled by
The car in 15 minutes (with relative speed of 20 km/hr.)
$\frac{15}{60} \times 20 = 5$km

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2001

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

ANCIENT HISTORY

1. Which one of the Chola kings conquered Ceylon?

(a) Aditya I (b) Rajaraja I
(c) Rajendra (d) Vijayalaya

Ans. (c) Rajendra Chola I (1014-1044 AD) continued his father's (Rajaraja) policy of aggressive conquests and expansion. Half part of Ceylon was conquered by Rajaraja while Rajendra conquered the whole of Ceylon.

2. Who among the following presided over the Buddhist Council held during the reign of Kanishka at Kashmir?

(a) Parsva (b) Nagarjuna
(c) Sudraka (d) Vasumitra

Ans. (d) : The fourth Buddhist council was held in 72 AD at Kundalvana, Kashmir. It was presided by Vasumitra, while Asvaghosa was his deputy. The council was held under the patronage of Kushan King Kanishka. Buddhism was divided into two sects namely Mahayana and Hinayana in that council.

3. Which one of the following animals was NOT represented on the seals and terracotta art of the Harappan culture?

(a) Cow (b) Elephant
(c) Rhinoceros (d) Tiger

Ans. (a) : From the seals, terracotta figurines, and copper tablets we get an idea on the religious life of the Harappans. The chief male deity was Pasupati, (proto-Siva) represented in seals as sitting in a yogic posture with three faces and two horns. He is surrounded by four animals (elephant, tiger, rhino, and buffalo each facing a different direction). Two deer appear on his feet.

4. Hoysala monuments are found in:

(a) hampi & Hospet
(b) Halebid & Belur
(c) Mysore & Bangalore
(d) Sringeri & Dharwar

Ans. (b) : It is the largest monument in Halebidu. The temple was built on the banks of a large man-made lake, and sponsored by King Vishnuvardhana of the Hoysala Empire. Its construction started around 1121 CE and was complete in 1160 CE.
Hoysala temples have features of both open (outer mantapa) and closed mantapa (innner mantapa). The ceilings of the mantapa are highly ornate bearing mythological figures and floral design.

5. Match List I with List II and select the correct answer using the code given below the lists:

List I (Bhakti Saint)	List II (Profession)
A. Namdev	1. Barber
B. Kabir	2. Weaver
C. Ravidas	3. Tailor
D. Sena	4. Cobbler

	A	B	C	D
(a)	2	3	1	4
(b)	3	2	4	1
(c)	3	2	1	4
(d)	2	3	4	1

Ans. (b) : Kabir, Namdev, Ravidas, Sena were the saints of Bhakti movement.
Namdev - He was a tailor from a lower caste family. He was the great devotee of Lord Vitthala of Pandharpur. He belonged to Varakari Sect.
Kabir - He was Weaver. He was brought up by a Muslim couple. His songs and couplets are known as Doha, Sakhi or Sabad.
Ravidas - He followed the path of Nirguna Brahma. He was a cobbler and founder of Ramdasi Sampradaya. He was the disciple of Saint Ramananda.
Sena- He was a barber and belonged to the Varkari sect dedicated to Lord Vithoba. He was also the disciple of Saint Ramananda.

6. Which one of the following pairs is correctly matched?

(a) Harappan Civilization : Painted Grey Ware
(b) The Kushans : Gandhara School of Art
(c) The Mughals : Ajantha Paintings
(d) The Marathas : Pahari School of Painting

Ans. (b) : Painted Greyware belonged to later Vedic period (1000–600BC). Ajanta paintings belong to the Gupta period. Pahari School came into existence during the Mughal period. Hence only option (b) is correct.